



Machine learning based on the multimodal connectome can predict the preclinical stage of Alzheimer's disease: a preliminary study

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Abstracts

Objectives Subjective cognitive decline (SCD) may be a preclinical stage of Alzheimer's disease (AD). Neuroimaging studies suggest that abnormal brain connectivity plays an important role in the pathophysiology of SCD. However, most previous studies focused on single modalities only. Multimodal combinations can more effectively utilize various information and little is known about their diagnostic value in SCD.

Methods One hundred ten SCD individuals and well-matched healthy controls (HCs) were recruited in this study (the primary sample: 35 SCD and 36 HC; the validation sample: 21 SCD and 18 HC). Multimodal imaging data were used to construct functional, anatomical, and morphological networks, respectively. These networks were used in combination with a multiple kernel learning-support vector machine to predict SCD individuals. We validated our model on another independent sample. Multiple linear regression (MLR) analyses were conducted to investigate the relationships among network metrics, cognition, and pathological biomarkers.

Results We found that the characteristics identified from the multimodal network were primarily located in the default mode network (DMN) and salience network (SN), achieving an accuracy of 88.73% (an accuracy of 79.49% for an independent sample) based on the integration of the three modalities. MLR analyses showed that increased AV45 SUVRs were significantly associated with impaired memory function, the enhanced functional connectivity, and the decreased morphological connectivity.

Conclusion This study suggests that abnormal multimodal connections within DMN and SN can be used as effective biomarkers to identify SCD and provide insight into understanding the pathophysiological mechanisms underlying SCD.

Key Points

- Multimodal brain networks improve the detection accuracy of SCD.
- Abnormal connections within DMN and SN can be used as effective biomarkers for the identification of SCD.

^aData used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

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The abbreviations of 246 ROIs are shown in Supplemental Table 1.

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Keywords Cognitive decline · Multimodal imaging · Machine learning · Default mode network · Linear models

Abbreviations

A β	Amyloid- β
AD	Alzheimer's disease
ADNI	Alzheimer's Disease Neuroimaging Initiative
APOE	Apolipoprotein E
CAT	Computational Anatomy Toolbox
CCI	Cognitive Change Index
CDR	Clinical dementia rating
CDT	Clock-Drawing Test
CEN	Central executive network
CI	Confidence intervals
CSF	Cerebrospinal fluid
DELCODE	DZNE-Longitudinal Cognitive Impairment and Dementia
DMN	Default mode network
DPABI	Data Processing & Analysis for Brain Imaging
DTI	Diffusion tensor imaging
DWI	Diffusion-weighted imaging
FA	Fractional anisotropy
FDR	False discovery rate
fMRI	Resting-state functional magnetic resonance imaging
GDS	Geriatric depression scale
GM	Gray matter
HC	Healthy controls
JSS	Jensen-Shannon distance-based similarity
LOOCV	Leave-one-out cross-validation
MCI	Mild cognitive impairment
MKL-SVM	Multiple kernel learning SVM
MMSE	Mini-Mental State Examination
MNI	Montreal Neurological Institute
PANDA	Pipeline for Analyzing brain Diffusion imAges
PET	Positron emission tomography
RAVLT	Rey Auditory Verbal Learning Test
ROC	Receiver operating characteristic curve
ROIs	Regions of interest
SCD	Subjective cognitive decline
SD	Standard deviation
sMRI	Structural MRI
SN	Salience network
SPM	Statistical parametric mapping
SUVRs	Standard uptake value ratios
SVM	Support vector machine
TE	Echo time
TI	Inversion time
TMT	Trail-Making Test
TR	Repetition time
WM	White matter
WMS-LM	Wechsler Memory Scale-Logical Memory

Introduction

Alzheimer's disease (AD) is a devastating neurodegenerative disease with severe and growing impact on the elderly health in modern societies [1]. In particular, AD has been found to cause numerous subtle changes in brain structure and function several years before the onset of overt cognitive impairment and clinical deficits [2]. Therefore, the early detection of brain disorganization in individuals likely to convert to AD is considered among the greatest challenges in current AD-related studies [3].

Subjective cognitive decline (SCD) may be a putative, preclinical asymptomatic stage of AD marked by self-reported deterioration in cognitive functioning, without overt deficit [3]. Despite the fact that SCD is considered a preclinical stage with high predictive value for subsequent cognitive degeneration, until now, the diagnosis of SCD is largely based on self-perceived symptoms in absence of any objective evidence [4, 5]. Non-invasive in vivo neuroimaging techniques have provided a feasible approach for identifying early characteristic changes in the brain, which may serve as potential biomarkers of SCD and provide guidance for identifying novel therapeutic targets [6].

Nevertheless, most previous studies on SCD focused on single modality only [7–11]. Given the limitations of single-modality studies, multimodal imaging analysis may provide a new approach for investigating diseases that affect multiple aspects of the brain structure and function. To date, multimodal neuroimaging techniques have rarely been used in SCD individuals. In recent years, machine-learning methods combining neuroimaging characteristics have been used in the early diagnosis of AD, which have tremendous potential for individual-based diagnosis [12–16]. However, it remains to be established, whether a machine-learning method combining multimodal neuroimaging techniques plays important roles in the early identification of SCD individuals.

In this study, we combined machine-learning techniques with multimodal neuroimaging to identify individuals with SCD from HC. We aimed to determine whether fusion of multimodal imaging data improved the classification performance and whether there were any associations between pathological biomarkers and neuroimaging features. This study may provide insight into understanding the pathophysiological mechanisms underlying SCD and provide potential quantitative neuroimaging biomarkers for SCD diagnosis.

Table 1 Demographic and neuropsychological data. Demographic and neuropsychological data. No significant differences were found in the demographic data, APOE genotype ($\epsilon 4$ carriers and non-carriers), CSF biomarkers, AV45 SUVRs, brain tissue volumes, and cognitive performance between the HC group and the SCD group. The SCD group displayed significantly increased CCI scores compared with HC group ($p < 0.001$)

Items	HC (n = 36)	SCD (n = 35)	Statistical value	p value
Age (years)	73.19 $\pm$ 7.03	72.04 $\pm$ 6.52	0.72	0.48 ^b
Education (years)	16.67 $\pm$ 1.97	17.23 $\pm$ 2.26	-1.12	0.27 ^b
Gender (male/female)	22/14	21/14	0.01	0.92 ^a
APOE $\epsilon 4$ (-/+)	31/36 (24/7)	26/35 (16/10)	1.70	0.19 ^a
CSF A β_{1-42} (pg/mL)	0/11/36 (1326.25 $\pm$ 507.76)	0/4/35 (1327.55 $\pm$ 313.11)	-0.01	0.99 ^b
CSF t-tau (pg/mL)	11/36 (235.27 $\pm$ 91.33)	4/35 (165.55 $\pm$ 39.84)	1.45	0.17 ^b
CSF p-tau (pg/mL)	11/36 (20.88 $\pm$ 8.27)	4/35 (14.43 $\pm$ 3.27)	1.49	0.16 ^b
[¹⁸ F] AV45 SUVRs	13/28/36 (1.18 $\pm$ 0.19)	12/18/35 (1.12 $\pm$ 0.11)	1.20	0.24 ^b
Intracranial volume (cm ³)	1417.85 $\pm$ 203.58	1414.85 $\pm$ 123.95	0.08	0.94 ^b
Gray matter volume (cm ³)	598.07 $\pm$ 69.93	597.38 $\pm$ 39.17	0.05	0.96 ^b
White matter volume (cm ³)	531.06 $\pm$ 89.32	520.69 $\pm$ 61.19	0.57	0.57 ^b
Ventricular volume (cm ³)	288.72 $\pm$ 65.43	296.78 $\pm$ 55.06	-0.56	0.58 ^b
Hippocampal volume (cm ³)	8.85 $\pm$ 1.08	8.85 $\pm$ 1.02	0.01	0.99 ^b
Left hippocampal volume (cm ³)	4.42 $\pm$ 0.53	4.42 $\pm$ 0.56	-0.01	0.99 ^b
Right hippocampal volume (cm ³)	4.43 $\pm$ 0.57	4.43 $\pm$ 0.49	0.03	0.98 ^b
Neuropsychiatric scores				
CCI	22/36 (13.86 $\pm$ 1.32)	35/35 (21.26 $\pm$ 5.37)	-6.32	< 0.001 ^{b*}
GDS-15	0.61 $\pm$ 0.80	0.91 $\pm$ 1.36	-1.15	0.26 ^b
General mental status				
MMSE	28.86 $\pm$ 1.50	28.63 $\pm$ 1.35	0.69	0.50 ^b
Memory function				
WMS-LM immediate recall	14.83 $\pm$ 3.59	14.97 $\pm$ 3.54	-0.16	0.87 ^b
WMS-LM delayed recall	13.92 $\pm$ 3.42	13.40 $\pm$ 3.79	0.60	0.55 ^b
RAVLT total	46.92 $\pm$ 9.33	45.29 $\pm$ 11.04	0.67	0.50 ^b
RAVLT immediate recall	9.39 $\pm$ 3.60	9.60 $\pm$ 3.27	-0.26	0.80 ^b
RAVLT delayed recall	8.61 $\pm$ 3.80	8.71 $\pm$ 4.72	-0.10	0.92 ^b
Attention				
TMT-A	29.83 $\pm$ 7.93	31.43 $\pm$ 8.10	-0.84	0.40 ^b
Executive function				
TMT-B	70.97 $\pm$ 33.06	79.17 $\pm$ 44.92	-0.88	0.38 ^b
Language ability				
Category fluency-animals	22.11 $\pm$ 6.21	22.83 $\pm$ 5.37	-0.52	0.61 ^b
Visuospatial function				
CDT	4.72 $\pm$ 0.51	4.74 $\pm$ 0.44	-0.18	0.86 ^b

Values are presented as the mean $\pm$ standard deviation (SD)

^a The p value was obtained by χ^2 test, ^b the p value was obtained by two-sample t tests

*indicates a statistical difference between groups, $p < 0.05$

Notably: CSF A β_{1-42} data (###) represents the number of subjects (A β_{1-42} positive/ with CSF data/ total subjects)

CSF t-tau data (##) represents the number of subjects (with CSF data/ total subjects)

CSF p-tau data (##) represents the number of subjects (with CSF data/ total subjects)

[¹⁸F] AV45 SUVRs data (###) represents the number of subjects (AV45 SUVRs positive/with CSF data/ total subjects)

CCI data (##) represents the number of subjects (with CCI data/ total subjects)

Abbreviations: HC health control, SCD subjective cognitive decline, APOE apolipoprotein E, CSF cerebrospinal fluid, SUVR standardized uptake values ratio, MMSE mini mental state examination, CCI Cognitive Change Index, GDS Geriatric Depression Scale, WMS-LM Wechsler Memory Scale Logical Memory, RAVLT Rey Auditory Verbal Learning Test, TMT Trail-Making Test, CDT Clock Drawing Test

Materials and methods

Alzheimer's Disease Neuroimaging Initiative

Data used in the preparation of this article were extracted from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (<http://adni.loni.usc.edu>). For more information, see the Supplementary Material and previous papers [17, 18]. Informed consent was obtained according to the Declaration of Helsinki.

Participants

We included 35 SCD subjects and 36 well-matched HCs from the ADNI database (ADNI-2 and ADNI-3 phase). The diagnostic criteria were described in the ADNI manual. The inclusion criteria for HC and SCD were described as previous study [19], and detailed information is included in the Supplementary Material.

Clinical and neuropsychological measurement

Demographic characteristics and neurocognitive performance data were downloaded from the ADNI database. For the primary analyses, all participants underwent a battery of cognitive evaluations, including global cognitive function, memory, attention, executive function, language ability, and visuospatial function.

Apolipoprotein E genotyping

Apolipoprotein E (APOE) genotypes of participants in this study were obtained from the ADNI database. All participants were classified as APOE ϵ 4 carriers and APOE ϵ 4 non-carriers. Notably, not all participants had APOE genotype data and detailed information is shown in Table 1.

Cerebrospinal fluid biomarkers

Lumbar puncture and cerebrospinal fluid (CSF) sample preparation were performed as described in the ADNI manual. CSF A β ₁₋₄₂, t-tau, and p-tau were measured using INNOBIA AlzBio3 immunoassay kit-based reagents. Notably, not all participants had CSF sample data since lumbar puncture is an invasive operation. In this study, 4 out of 35 SCD subjects and 11 out of 36 HCs had CSF sample data available (Table 1).

[¹⁸F] AV45 positron emission tomography scans

[¹⁸F] AV45 positron emission tomography (PET) data were processed as described in the standardized protocol. Mean florbetapir standard uptake value ratios (SUVRs) were

computed within these brain regions (lateral and medial anterior frontal, lateral temporal, posterior cingulate, and lateral parietal cortex) and normalized to the whole cerebellum as the reference region. In this study, 18 out of 35 SCD individuals and 28 out of 36 HC had PET SUVR data available (Table 1).

MRI acquisition

All participants were examined on a SIEMENS 3.0-T scanner. The examination protocol included a high-resolution T1-weighted sequence, an rs-fMRI sequence and a diffusion-weighted imaging (DWI) sequence, which are shown in the Supplementary Material.

Image preprocessing and network construction

Functional network

The rs-fMRI data were preprocessed by the Data Processing & Analysis for Brain Imaging (DPABI V4.1, <http://rfmri.org/dpabi/>). The main preprocessing steps included slice time correction, head motion correction, normalization, filtering, and multiple linear regression analysis (including the Friston 24 parameters, cerebrospinal fluid, and white matter signals). In this study, the two groups did not show significant differences in the mean frame-wise displacement as suggested by Jenkinson et al [20]. To define the network nodes, the Human Brainnetome Atlas was utilized to divide the whole brain into 246 regions of interest (ROIs) (the abbreviations in Supplemental Table 1). To define network edge, we calculated Pearson correlation of the regional mean time series between each pair of 246 ROIs. To further remove spurious correlations, only those correlation coefficients whose corresponding *p* values lower than a statistical threshold (*p* < 0.05, false discovery rate [FDR]-corrected) were retained [21].

Morphological network

Brain tissue segmentation was performed using the Computational Anatomy Toolbox (CAT12, <http://www.neuro.uni-jena.de/cat/>) as implemented in the Statistical Parametric Mapping analysis package (SPM12, <http://www.fil.ion.ucl.ac.uk/spm/software/spm12/>). To construct morphological network, we utilized a Jensen-Shannon distance-based similarity (JSS) to quantify morphological connectivity between each pair of 246 ROIs [22]. The JSS between any pair of ROIs was calculated by their probability distribution function. The range of JSS value is from 0 to 1. Higher JSS value was obtained when the gray matter (GM) density distribution of two ROIs was closer.

Anatomical network

Diffusion data preprocessing and anatomical network construction were performed in the Pipeline for Analyzing brain Diffusion imAges (PANDA, <https://www.nitrc.org/projects/panda/>). The tractography was terminated if it turned an angle greater than 45° or reached a voxel with a fractional anisotropy (FA) less than 0.2. These 246 brain regions were used as the nodes to carry out tractography. For the weighted network, we defined the mean FA values of the connected fibers between 2 nodes as the weights of the edges. To reduce the risk of false-positive connections, two regions were considered structurally connected if at least 3 fibers were located in these 2 regions [23]. As a result, for each participant, the anatomical network was represented by a symmetric weighted 246×246 matrix.

Data mining

In our study, the LIBSVM toolbox for MATLAB was applied to implement the support vector machine (SVM) classification. Dimensionality reduction was necessary prior to discriminant analysis. Consequently, rank-sum tests comparing every connection across the classes were performed separately for the training set, yielding a p value for each connection. Statistically significant features ($p < 0.001$, uncorrected) were selected as in the previous study, whereas the rest were eliminated [24, 25]. Then, the discriminant analysis was performed by using an SVM as a classifier. To further integrate the complementary information from different modalities, we employed the multiple kernel learning SVM (MKL-SVM) to fuse these features as described in the other studies [26, 27]. We used an inner leave-one-out cross-validation (LOOCV) method to determine the optimal parameters (including different kernel types: linear, $t = 0$; polynomial, $t = 1$; radial basis function, $t = 2$; and different trade-off parameters C : 0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000) and outer LOOCV was used to determine the classification performance as we did in our previous study. Due to the limited sample size, we adopted the LOOCV for validation, since the LOOCV can be more robust than K-fold or hold out. Meanwhile, the mechanism of SVM is based on the support vector, which is more suitable for the small sample size than other methods such as deep neural networks (DNNs). Moreover, this work is mainly focused on validation, the SVM is an attempt, and any other multi-view methods in machine learning can be adopted to combine the information from different modals. However, the LOOCV needs to train the model multiple times, which is extremely time-consuming. Finally, we had trained and summarized 7 models to compare the judgment effects of single modality and different modal combinations. The performance of a classifier could be quantified using accuracy, sensitivity, specificity, and the area under the receiver

operating characteristic curve (ROC). We also used contingency tables to compare the coherence of the SVM models across the different modalities according to the previous study [28].

Consensus features for each modality

When using the outer LOOCV method to evaluate classification performance, the feature set selected as the output was slightly different, and different features provided different contributions to classification. Consensus features refer to the connections that could be selected as the feature input in each cross-validation [29, 30]. Thus, we focused on consensus features for each modality.

Statistical analysis

Differences between the HC group and the SCD group in demographic, neuroimaging characteristics, and cognitive performance were assessed using two-sample t test or chi-square (χ^2) test using Statistical Package for Social Sciences (SPSS V22). The significance level was set at $p < 0.05$.

Multiple linear regression analyses were conducted to investigate the relations between AV45 SUVrs, connections, and cognition performance, while controlling for potential confounders (including age, gender, years of education, and APOE genotypes).

Furthermore, we conducted mediation analyses to test the primary hypothesis of whether altered connections mediated the relationships between AV45 SUVrs and cognition, controlling for covariates. We computed the bias-corrected 95% confidence intervals (CIs) for the size of the mediating effects with bootstrapping ($k = 10000$ samples). The mediating effect is said to be present if the 95% CI does not contain zero. Mediation analyses were conducted in PROCESS (<http://www.afhayes.com/>) for the SPSS framework.

Validation sample

Next, we performed external validation of our models on a sample comprising HC ($n = 18$) and SCD ($n = 21$) from ADNI database (ADNI-2 and ADNI-3 phase; <http://adni.loni.usc.edu>). The detailed information is provided in the Supplementary Material.

Results

Demographics and clinical characteristics

Demographic and clinical data for the HC group and SCD group are shown in Table 1. No significant differences were

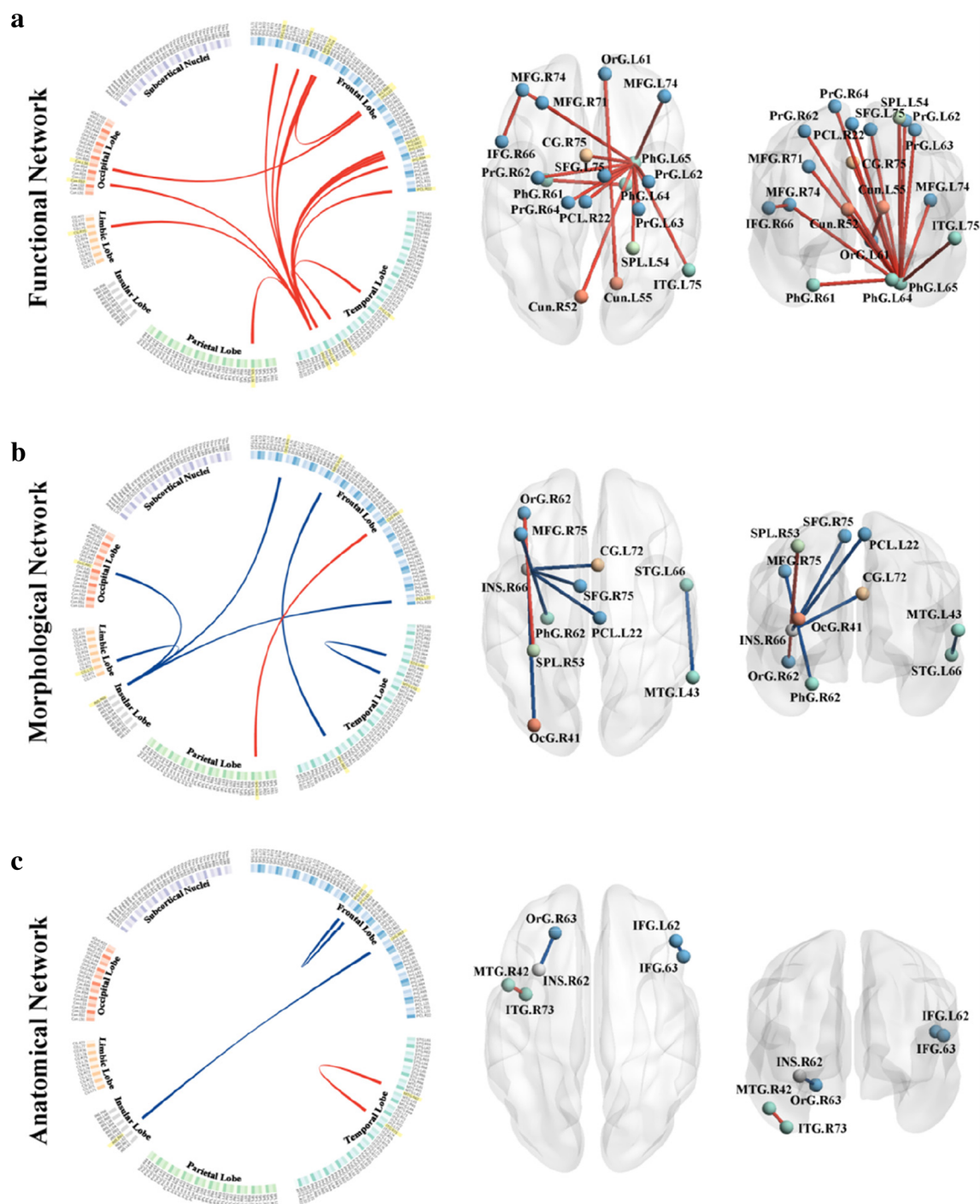


Fig. 1 The “consensus” features identified via the different modalities. **A** Functional network, **B** morphological network, and **C** anatomical network. Note: red lines represent higher connections in SCD than HC;

blue lines represent lower connections in SCD than HC. Abbreviations: SCD, subjective cognitive decline; HC, healthy controls; the abbreviations of 246 brain regions could be seen in Supplemental Table 1

found in the demographic data, APOE genotype ($\epsilon 4$ carriers and non-carriers), CSF biomarkers, AV45 SUVrs, and the brain tissue volumes between the HC group and SCD group. The SCD group displayed significantly increased CCI scores compared with the HC group ($p < 0.001$). Additionally, there were no significant differences in cognitive performance.

Consensus features

We examined the consensus connections for the classification model that combined the three modal networks (Fig. 1 and Supplemental Table 2). The functional network yielded 16 such connections, which exhibited higher

ROC curves of classification results using the “connection” features

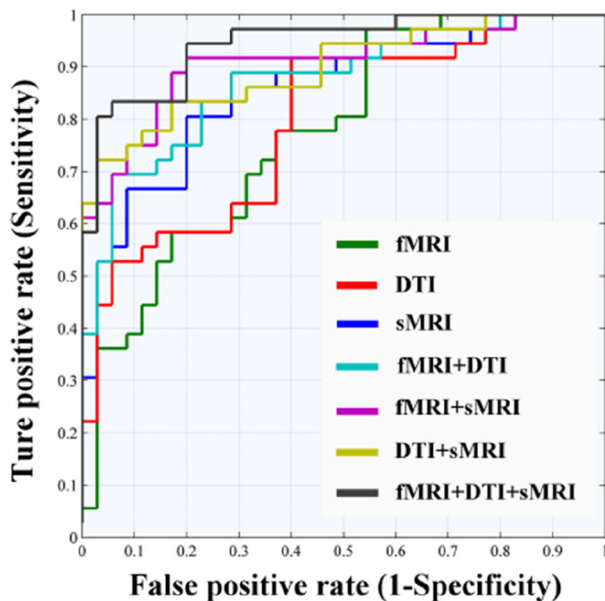


Fig. 2 Discrimination accuracy of different models. The classification accuracy improved after combining the network features of the three modalities, achieving the accuracy of up to 88.73% with a sensitivity and specificity of 80.56% and 97.14%, respectively. The detailed data were provided in Table 2. Abbreviations: ROC, receiver operating characteristic curve; fMRI, functional magnetic resonance imaging; DTI, diffusion tensor imaging; sMRI, structural magnetic resonance imaging

connection strength in the SCD group than in the HC group ($p < 0.001$, uncorrected) (Fig. 1A and Supplemental Table 2). These increased connections were comprised mainly of inter-region connections, which linked the left parahippocampal gyrus (PhG) with the

frontal lobe (SFG, MFG, PrG), temporal lobe (PhG, ITG), parietal lobe (PCL), occipital lobe (Cun), and limbic lobe (CG) (14/16, 87.5%). The morphological network yielded 7 connections: 6 of these connections were weakened in SCD, especially those between the right insular gyrus (INS) and the frontal lobe (SFG), parietal lobe (PCL), occipital lobe (OcG), and limbic lobe (CG) (4/6, 66.67%); 1 of the connections was enhanced in SCD, which linked right orbital gyrus (OrG) and right superior parietal lobule (SPL) (Fig. 1B and Supplemental Table 2). The anatomical network yielded 3 connections: 2 of these connections were decreased in SCD, which were between right OrG and INS, and within the left inferior frontal gyrus (IFG); 1 of the connections was increased in SCD, which linked right ITG and right middle temporal gyrus (MTG) (Fig. 1C and Supplemental Table 2).

Discriminative analysis and validation in the independent sample

In this study, MKL-SVM was used to classify whether a sample belonged to the SCD group or the HC group (Fig. 2, Table 2). For single-modality analyses, the connections of the morphological network exhibited the highest accuracy rate (78.87%), followed by the functional network which achieved the accuracy rate of 67.61%. Typically, classification accuracy improved after combining the network features of the three modalities, achieving the accuracy of up to 88.73% with a sensitivity and specificity of 80.56% and 97.14%, respectively. In this study, we performed external validation of our classification models on an independent sample. We used the consensus connections obtained from the discovery dataset as the selected features. Demographic and clinical data for the HC group

Table 2 Results of the discrimination analyses derived from the connections between HC and SCD in the discovery and validation dataset

Feature	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
Discovery dataset				
fMRI	67.61	69.44	65.71	0.77
DTI	64.79	58.33	71.43	0.80
sMRI	78.87	77.78	80.00	0.86
fMRI + DTI	80.28	83.33	77.14	0.87
fMRI + sMRI	83.10	83.33	82.86	0.91
DTI + sMRI	81.69	75.00	88.56	0.90
fMRI + DTI + sMRI	88.73	80.56	97.14	0.95
Validation dataset				
fMRI + DTI + sMRI	79.49	61.11	71.43	0.72

Abbreviation: HC health control, SCD subjective cognitive decline, AUC the area under the curve, DTI diffusion tensor imaging, fMRI functional magnetic resonance imaging, sMRI structural magnetic resonance imaging

Table 3 Multiple linear regression models and mediation analyses for AV45 SUVRs, connections, and cognition

	RAVLT immediate recall		RAVLT delayed recall		MC (SFG.R75-INS.R66)		FC (PhG.L65-ITG.L75)	
	β	<i>p</i>	β	<i>p</i>	β	<i>p</i>	β	<i>p</i>
Model 1								
AV45 SUVRs	-0.56 (-1.05-0.08)	0.026*	-0.46 (-0.79-1.31)	0.01*	-0.62 (-1.10-0.13)	0.017*	0.56 (0.11 1.02)	0.02*
MC (SFG.R75-INS.R66)	0.45 (-0.09 0.98)	0.095	0.25 (-0.16 0.66)	0.213	—	—	—	—
FC (PhG.L65-ITG.L75)	-0.36 (-0.97 0.26)	0.230	-0.25 (-0.70 0.20)	0.248	—	—	—	—
Model 2								
AV45 SUVRs	-0.47 (-1.12 0.17)	0.137	-0.5 (-0.94-0.06)	0.03*	—	—	—	—
Model 3								
AV45 SUVRs	-0.58 (-1.23 0.06)	0.073	-0.51 (-0.94-0.08)	0.025*	—	—	—	—
Model 4								
MC (SFG.R75-INS.R66)	0.15 (-0.50 0.80)	0.622	-0.07 (-0.51 0.38)	0.753	—	—	—	—
FC (PhG.L65-ITG.L75)	0.03 (-0.67 0.74)	0.920	0.09 (-0.38 0.56)	0.679	—	—	—	—

Displayed are standardized regression coefficients (β 95% confidence interval) (with parenthetical *p* values)

Model 1 controlled for age, gender, years of education, and APOE genotypes

Model 2 controlled for age, gender, years of education, APOE genotypes, and MC (SFG.R75-INS.R66)

Model 3 controlled for age, gender, years of education, APOE genotypes, and FC (PhG.L65-ITG.L75)

Model 4 controlled for age, gender, years of education, APOE genotypes, and AV45 SUVRs* indicates statistically significant, $p < 0.05$ And they were shown in bold for emphasis

Abbreviations: SUVR standardized uptake values ratio, RAVLT Rey Auditory Verbal Learning Test, MC morphological connectivity, FC functional connectivity, SFG superior frontal gyrus, INS insular gyrus, PhG parahippocampal gyrus, ITG inferior temporal gyrus

and SCD group were summarized in Supplementary Table 3. Combining the network features of the three modalities achieved the accuracy of 79.49% (Table 2). We

also used contingency tables to compare the coherence of the SVM models across the different modalities (detailed information in Supplemental Table 4).

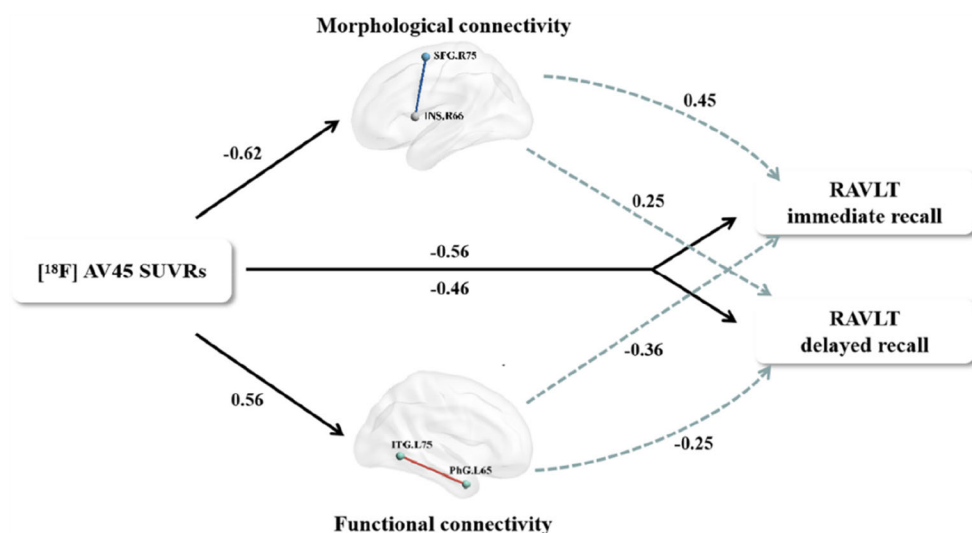


Fig. 3 Relationships between biomarkers, altered connections, and cognition in SCD. The AV45 SUVRs were significantly associated with RAVLT immediate recall ($\beta = -0.56$, $p = 0.026$), RAVLT delayed recall ($\beta = -0.46$, $p = 0.010$), the morphological connectivity between SFG.R75 and INS.R66 ($\beta = -0.62$, $p = 0.017$), and the functional connectivity between PhG.L65 and ITG.L75 ($\beta = 0.56$, $p =$

0.020) while controlling for potential confounders (including age, gender, years of education, and APOE genotypes). Abbreviations: SCD, subjective cognitive decline; RAVLT, Rey Auditory Verbal Learning Test; SUVRs, standard uptake value ratios; SFG, superior frontal gyrus; INS, insular gyrus; PhG, parahippocampal gyrus; ITG, inferior temporal gyrus; APOE, apolipoprotein E

Relationships between biomarkers, altered connections, and cognition performance

We performed multiple linear regression analyses to investigate the relationships between AV45 SUVRs, consensus features, and cognitive performance in the SCD group (Fig. 3, Table 3). We found that AV45 SUVRs were significantly associated with RAVLT immediate recall ($\beta = -0.56$, $p = 0.026$), RAVLT delayed recall ($\beta = -0.46$, $p = 0.010$), the morphological connectivity between SFG.R75 and INS.R66 ($\beta = -0.62$, $p = 0.017$), and the functional connectivity between PhG.L65 and ITG.L75 ($\beta = 0.56$, $p = 0.020$) while controlling for potential confounders (including age, gender, years of education, and APOE genotypes). The relationship of AV45 SUVRs and RAVLT delayed recall remained significant while controlling for age, gender, years of education, APOE genotypes, and morphological connectivity (SFG.R75-INS.R66) or functional connectivity (PhG.L65-ITG.L75). However, the indirect effect of the morphological connectivity (SFG.R75-INS.R66) and the functional connectivity (PhG.L65-ITG.L75) had no statistical significance. Correlations between AV45 SUVRs and other cognitive assessments, and consensus connections, are also shown in Supplementary Table 5.

Discussion

In this study, we utilized multimodal neuroimaging data to construct functional, morphological, and anatomical networks at the individual level, to identify neuroimaging markers in subjects with SCD. The characteristics identified from the multimodal network were primarily located in the default mode network (DMN) and salience network (SN), achieving an accuracy of 88.73% (79.49% on an independent sample). Relative to any single modality, integration of the three modalities yields more accurate predictions. Moreover, some neuroimaging biomarkers were related to pathological biomarkers, which might mediate the association between pathological characteristics and memory function.

SCD may be a stage of early neurodegeneration but with clinically normal cognition contributing to functional preservation or compensation [31, 32]. The functional connections extracted as features for classification were mainly between the left PhG and other DMN-related regions, and all of them were enhanced in individuals with SCD compared to the HC. The DMN involved in memory and self-related processing is considered the most vulnerable functional brain regions in AD [33, 34]. Previous studies proposed that cognitive complaints, which were the primary performance of SCD individuals, were associated with the cortical atrophy of PhG affected early by AD pathology [35–38]. Recently, Viviano et al summarized previous researches and proposed a functional model

where the functional connectivity in the DMN showed a non-linear change trajectory over the progression from subjective to objective cognitive decline in SCD individuals [6]. Early synaptic disruption may lead to compensatory recruitment that underlie increased functional connectivity which may transform to decreased functional connectivity with neurodegeneration progression and objective cognitive impairment becomes increasingly apparent [6]. Therefore, we speculate that the higher functional connections within the DMN may represent a compensatory mechanism to maintain stable cognitive ability in SCD individuals.

For single-modality analyses, the connections in the morphological network were used as features for classification, exhibiting a diagnostic accuracy rate of 78.87%. This finding indicated that morphological network analysis at the individual level might have the potential diagnostic value in identifying SCD individuals. In addition, Tong et al proposed a novel grading biomarker based on the morphological imaging and demonstrated a significant contribution in accurate prediction of conversion from MCI to AD [39]. In our study, the majority of abnormal morphological connections linked the right INS and other regions, and all of them were decreased in individuals with SCD compared to the HC. Menon proposed a triple network model, which highlighted the crucial role of the right INS, for initiating the network switching leading to the subsequent engagement of the central executive network (CEN) and the disengagement of the DMN [40–42]. According to this model, we deduced that enhanced functional connectivity in the DMN may be caused by decreased morphological connectivity in the SN for individuals with SCD.

Postmortem evidence has suggested that white matter abnormalities may be related to Wallerian-like degeneration triggered by the deposition of cortical pathology in AD, whereas white matter damages in the normal elderly are due to ischemia [43, 44]. At the preclinical stage of AD, our study indicated that the anatomical connections altered in SCD individuals were located in the DMN (the connections between right ITG and MTG and within the left IFG) and SN (the connection between right OrG and INS). This finding is consistent with the previous study that Yan et al identified the most discriminating anatomical connections, which were mainly distributed in the DMN and subcortical structures [45]. The prospective studies on large populations are required to further investigate the anatomic-connectome-based biomarkers for the early detection of SCD individuals.

After merging the network connections of the three modalities (fMRI, DTI, and sMRI), the diagnostic accuracy of up to 88.73% was higher than that obtained using any single modality, indicating that the integration of multimodal features could provide complementary information carried by different mode networks. Although the connections identified by each modality were different, the

convergent pattern was that all of them were mostly located within the DMN and SN. Recently, Yan et al employed functional and anatomical connectivity to differentiate SCD individuals from HC. Compared to the classification performance of single-modal methods (fMRI: 77.81%; DTI: 58.51%), the integration of fMRI and DTI exhibited the higher accuracy rate (80.24%) [45]. In our study, we applied the MKL-SVM method combined with multimodal brain networks and further improved the diagnostic accuracy (88.73%). This finding suggested that multimodal abnormalities in the DMN and SN could be used as biomarkers of SCD. We performed external validation of our classification models on an independent sample. Combining the network features of the three modalities achieved the accuracy of 79.49%. The small sample size may be an important factor of limited reproducibility. In addition, the selectivity of model algorithm also affect the generalization capability. Due to the limited sample size, we adopted the LOOCV for validation, since the LOOCV can be more robust than K-fold or hold out. Meanwhile, the mechanism of SVM is based on the support vector, which is more suitable for the small sample size than other methods such as deep neural networks (DNNs). Moreover, the SVM is an attempt and multi-view method in machine learning, which can be adopted to combine the information from different modals.

Additionally, we conducted mediation analyses to explore whether the connections identified by each modality mediated the relationships between AD pathology (AV45 SUVRs) and cognition in SCD subjects. The DZNE-Longitudinal Cognitive Impairment and Dementia (DELCODE) study demonstrated that lower $A\beta_{1-42}$ levels of CSF in SCD individuals were closely related to the perceived decline in memory performance, consistent with our result [46, 47]. Furthermore, a 3-year follow-up study indicated that DMN and SN connectivity was synergistic with $A\beta$ burden in predicting decline in preclinical Alzheimer cognitive composite performance [48]. According to previous evidence, we speculated that abnormal connections within DMN and SN might mediate the association between pathological characteristics and memory function in SCD individuals.

Although our study tried to provide a new approach for the early identification of SCD, a few limitations still require future study. First, the pattern of the multimodal brain network is affected by different parcellation strategies. Other brain-wide graphs may be used to further assess the reliability in the differentiation of SCD individuals. Second, this study was performed on a small sample size of SCD individuals with pathological markers, and we look forward to expanding the sample size to validate our results in future studies. Third, we compared only

SCD and HC individuals; we did not involve MCI and AD patients. We added those participants in the subsequent study.

Conclusion

We applied the MKL-SVM method combined with multimodal connectome information to distinguish SCD individuals from HC. Relative to any single modality, integration of the three modalities yields more accurate predictions, suggesting that multimodal neuroimaging data could be effectively utilized to obtain complementary information regarding different modalities. Furthermore, these findings provide direct evidence that the abnormal connections within DMN and SN could be used as effective biomarkers for the early-stage diagnosis of AD.

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Declarations

Guarantor The scientific guarantor of this publication is Prof. Feng Bai.

Conflict of interest The authors of this manuscript declare no relationships with any companies whose products or services may be related to the subject matter of the article.

Statistics and biometry No complex statistical methods were necessary for this paper.

Informed consent Subject data used in the preparation for this article was obtained from publicly available study samples, i.e., ADNI. Written informed consent was obtained as part of these studies.

Ethical approval Institutional Review Board approval was obtained.

Study subjects or cohorts overlap Subject data (SCD patients) used in the preparation for this article was obtained from publicly available study samples, i.e., ADNI, which have not been used in previous publications.

Methodology

- Retrospective
- Cross sectional study
- Multicenter study

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